

Basics

In essence this is **Physics Basics 101** :

Base Quantities

These are all known as SI Units, from the French International System of Units

The SI system defines 7 base quantities :

Base quantity	Symbol	Base unit	Symbol
Time	t	second	s
Length	l	meter	m
Mass	m	kilogram	kg
Electric current	I	ampere	A
Thermodynamic temperature	T	kelvin	K
Amount of substance	n	mole	mol

From these we can create **Derived Quantities**

Derived Quantities

All other quantities are derived quantities and are based on two or more of the **Base Quantities** , derived using an equation. Refer to the **Rules of Indices**

Derived Quantity Examples :

- Velocity
 - Formula : $\frac{\Delta l}{t}$
 - Δl has the units m , t has the units s
 - Therefore the units becomes $\frac{m}{s}$, which using indice rules, we turn into ms^{-1}

- Acceleration

- Formula : $\frac{\Delta v}{t}$
- Δv has the units ms^{-1} , t has the units s
- Therefore the units becomes $\frac{ms^{-1}}{s}$, then ms^{-2}

- Force

- Formula : $F = ma$
- m has the units kg , a has the units ms^{-2}
- Therefore the units becomes $kgms^{-2}$

- Pressure

- Formula : $\frac{F}{A}$
- F has the units $kgms^{-2}$, A has the units m^2
- Therefore the units becomes $\frac{kgms^{-2}}{m^2}$, then $kgm^{-1}s^{-2}$

- Work / Energy

- Formula : $W = Fd$
- F has the units $kgms^{-2}$, d has the units m
- Therefore the units becomes kgm^2s^{-2}

- Power

- Formula : $\frac{E}{t}$
- E has the units kgm^2s^{-2} , t has the units s
- Therefore the units becomes $\frac{kgm^2s^{-2}}{s}$, then kgm^2s^{-3}

- Voltage

- Formula : $\frac{P}{I}$
- P has the units kgm^2s^{-3} , I has the units A
- Therefore the units becomes $\frac{kgm^2s^{-3}}{A}$, then $kgm^2s^{-3}A^{-1}$

- Resistance

- Formula : $\frac{V}{I}$
- V has the units $kgm^2s^{-3}A^{-1}$, I has the units A
- Therefore the units becomes $\frac{kgm^2s^{-3}A^{-1}}{A}$, then $kgm^2s^{-3}A^{-2}$

